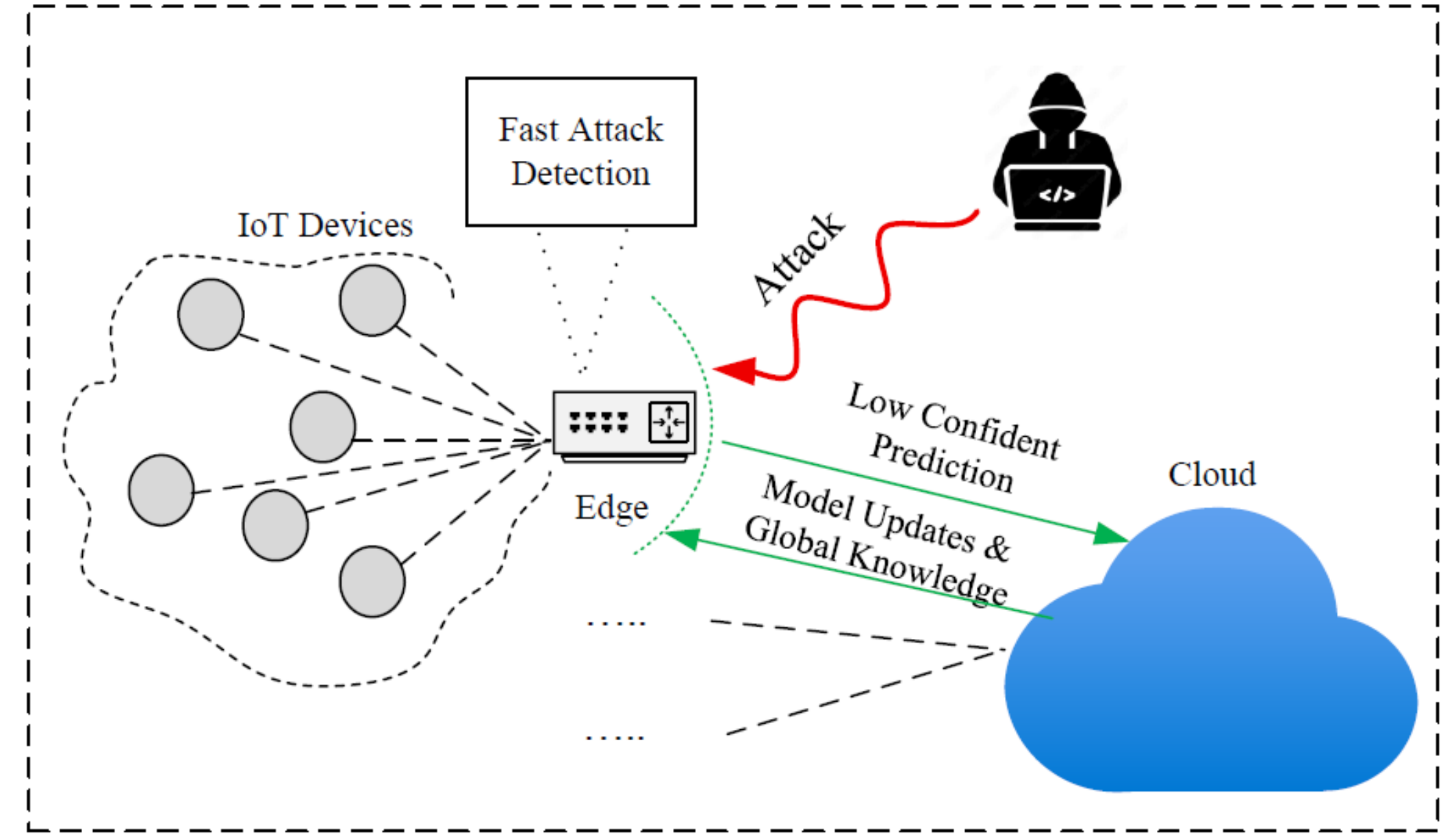


Introduction

- **Background:** IoT systems are ubiquitous but vulnerable. Traditional security fails against sophisticated attacks. Recent solutions use ML or DNN, but face a critical trade-off.
- **The Dilemma:** DNN-based IDS is accurate but too costly for IoT devices. Lightweight ML is fast but misses sophisticated attacks
- **Problems:** Current solutions sacrifice either accuracy or speed. IoT devices lack sufficient CPU, memory, and battery capacity to run complex models. Static IDS designs fail to adapt against evolving attacks and concept drift. High-volume attacks like DDoS overwhelm systems before detection mechanisms can respond.
- **The Gap:** No lightweight IDS that balances accuracy, speed, and adaptability

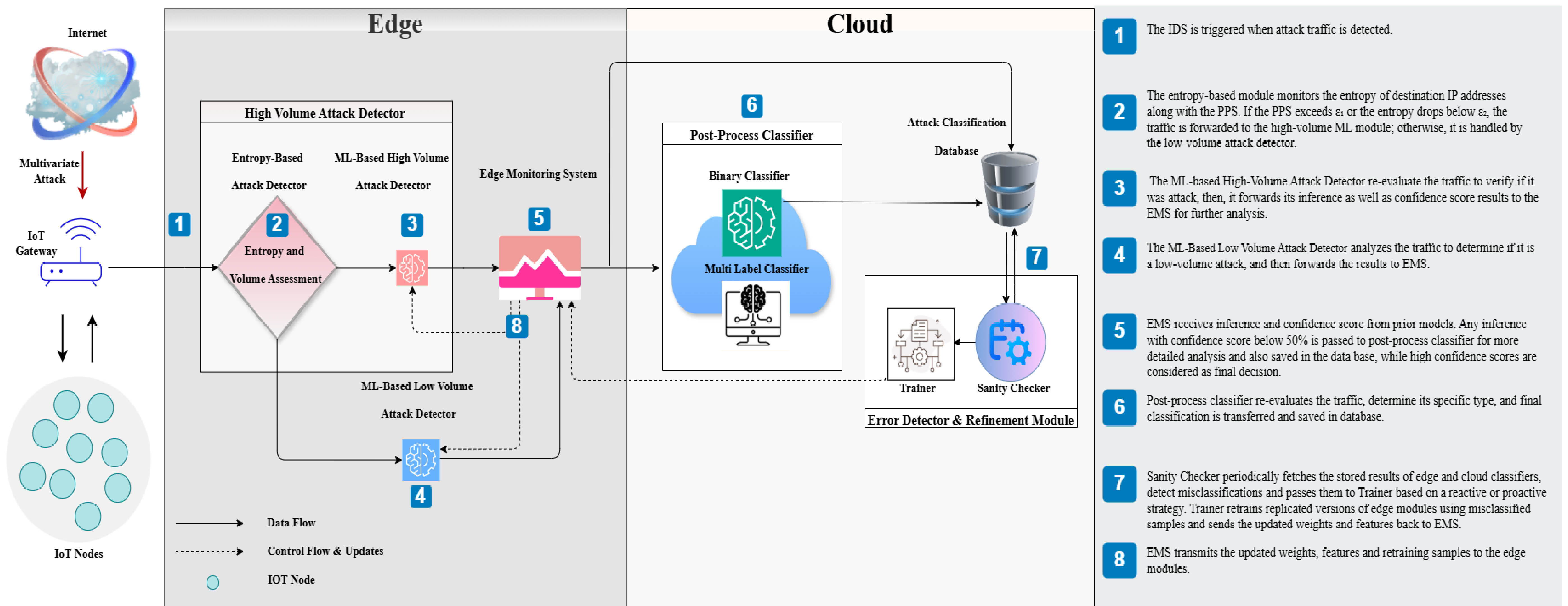
Motivation

Key Challenge: How can we design an IDS that achieves fast detection, high accuracy, and remains lightweight for IoT devices?



Methodology & Framework

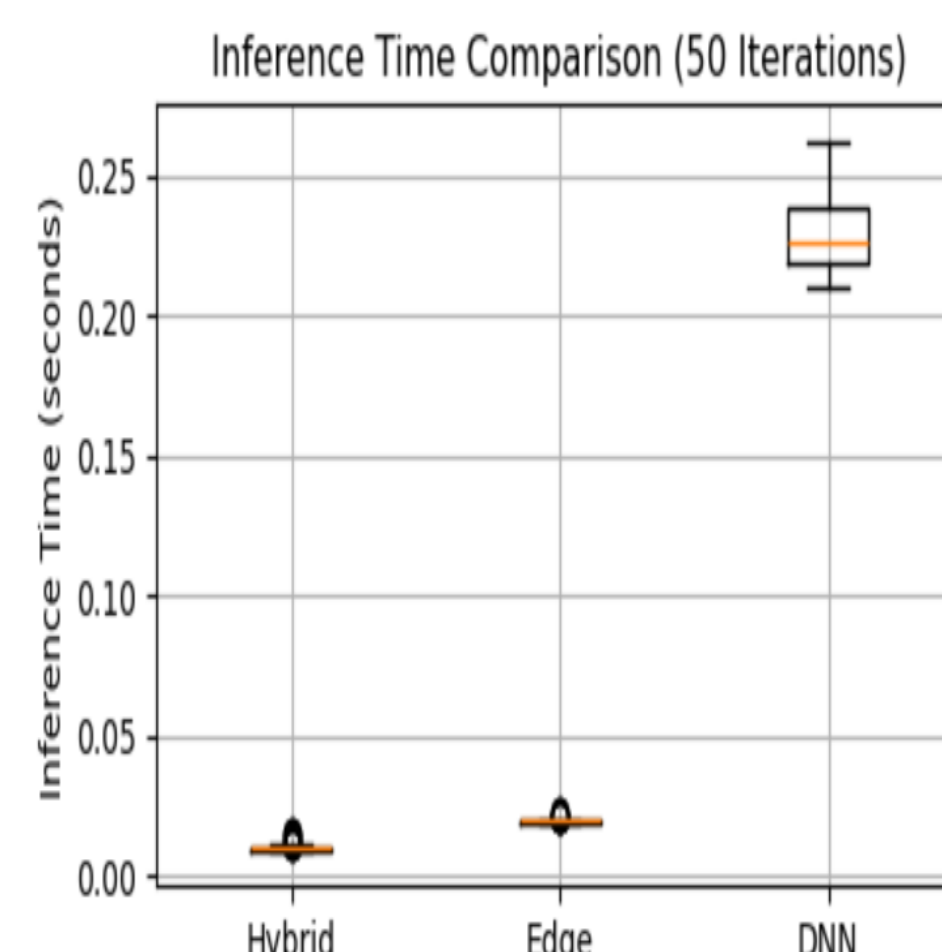
- **High-Volume Attack Detector (Two Stage):** Employs an entropy-based detector to identify DDoS/DoS attacks, followed by a lightweight ML-based classifier for secondary validation.
- **Low-Volume Attack Detector:** Employs a lightweight binary classifier (e.g., Decision Tree) to detect non-high-volume attack.
- **Edge Monitoring System:** Performs local retraining and confidence-based cloud offloading.
- **Post-Process Classifier:** High-accuracy reclassification of low-confidence edge predictions.
- **Error Detection & Refinement:** Monitors performance, detects drift, and automatically retrains models.



Performance Evaluation

TABLE I
EVALUATION METRICS FOR THE PROPOSED IDS AT VARIOUS
CONFIDENCE THRESHOLDS AND BASELINES

Framework / Confidence	Class 1			W. Avg.		
	P	R	F1	P	R	F1
Cloud Baseline	1.00	0.99	0.99	0.99	0.99	0.99
Edge Baseline	0.87	1.00	0.93	0.89	0.88	0.85
Proposed IDS (0.85)	0.92	1.00	0.96	0.94	0.93	0.93
Proposed IDS (0.95)	0.99	1.00	0.99	0.99	0.99	0.99



Conclusions

The proposed framework balances accuracy, speed, and resource efficiency for IoT security. It achieves perfect attack detection (recall = 1.0) with minimal false positives (precision = 0.92), while outperforming cloud-based approaches and existing baselines in response time. Its lightweight design supports deployment on edge devices, adapts to evolving threats, and remains resilient during network disruptions.

